

DiaTool-DPO

Multi-Turn Direct Preference Optimization for Tool-Augmented LLMs

Improving dialogue flow via automatic paired-trajectory construction without human labeling

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Missing Follow-ups

When user queries lack required arguments, SFT-trained models often skip slot-filling questions and invoke tools with incomplete parameters.

Hallucinated Calls

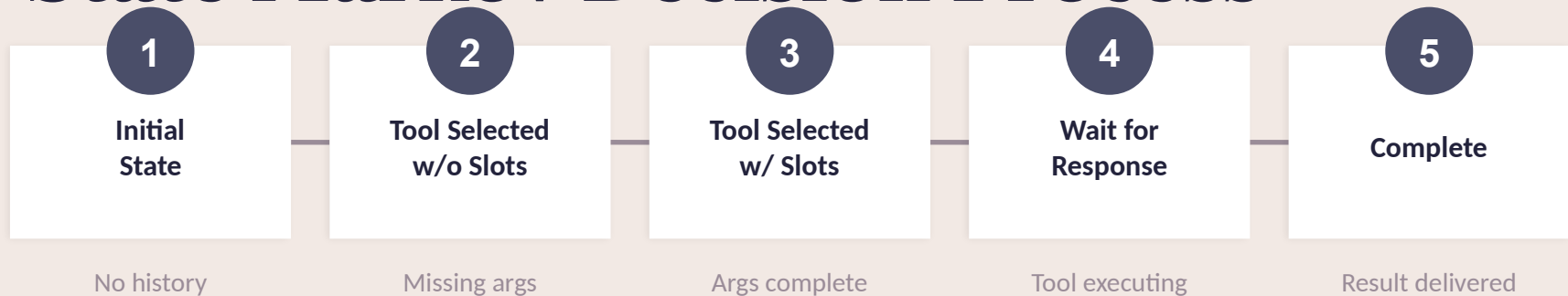
Without explicit rejection training, models generate tool calls for out-of-scope requests, producing fabricated invocations that degrade reliability.

Redundant QA Loops

Even when all arguments are present, models may unnecessarily enter slot-filling cycles, increasing latency and harming user experience.

DiaTool-DPO replaces static SFT imitation with preference-optimized trajectory learning.

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process



All five states are internal (unobserved). The model learns to transition between them based on dialogue context, tool schema, and slot completeness.

State diagram reference: figure_1.jpg (arXiv:2504.02882)

Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Description	Example
Type 1	$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$	Query contains all required arguments; no slot-filling needed.	"Book a flight from NYC to LAX on May 10."
Type 2	$1 \rightarrow (2 \times N) \rightarrow 3 \rightarrow 4 \rightarrow 5$	Query lacks some arguments; slot-filling QA repeats N times.	"Book a flight to LAX." $\rightarrow$ [Ask dates & origin] $\rightarrow$ call
Type 3	$1 \rightarrow 1$	Requested functionality is unavailable; tool call must be rejected.	"Translate this text to Klingon." $\rightarrow$ Reject (no tool)

Each trajectory class maps to a distinct dialogue control skill, enabling systematic preference-pair generation.

Automatic Paired Trajectory Construction Without Human Labels

Query	Chosen Traj.	Rejected Traj.	Learning Lesson	Count	Difficulty
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089	Easy
Type 1	1→3→4→5	1→(2×N)→3→4→5	Prevent redundant slot-filling	562	Hard
Type 1	1→3→4→5	1→(2×M)→3→4→5	Prevent redundant slot-filling	2,530	Hard
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089	Easy
Type 2	1→(2×N)→3→4→5	1→3→4→5	Prevent slot hallucination	562	Hard
Type 3	1→1	1→3→4	Tool call reject	567	Hard

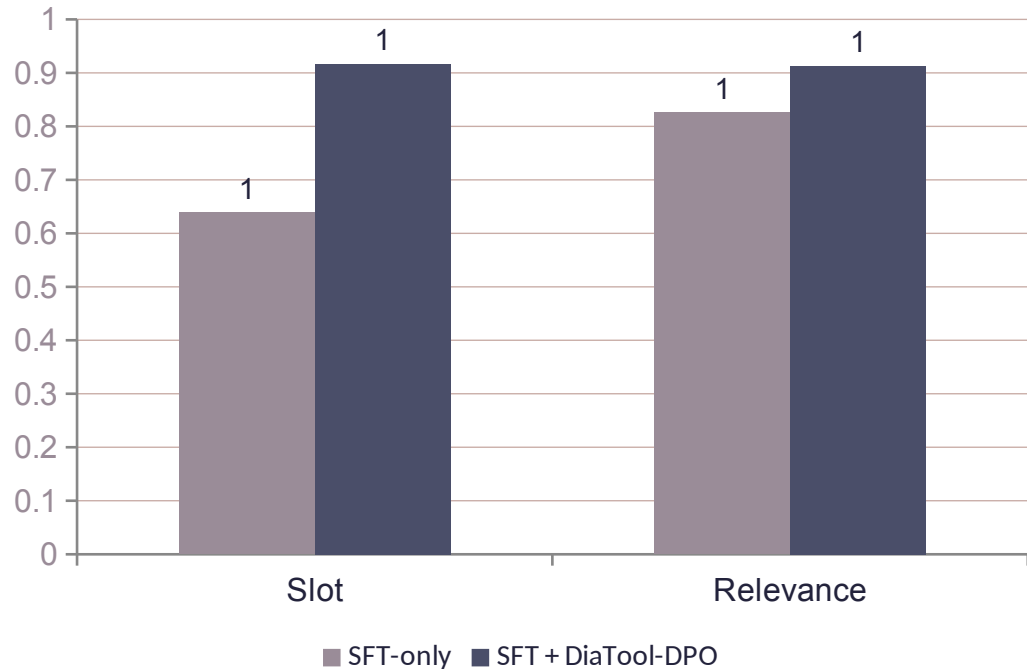
Selected representative pairs shown. Full table includes additional hard/easy variants and accept/reject combinations.

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Model	Method	Call	Completion	Slot	Relevance	Micro Avg	Macro Avg
Prop.-8B	DiaTool-DPO-only	0.314	0.700	0.833	0.609	0.575	0.614
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	SFT + DiaTool-DPO	0.886	0.929	0.833	0.826	0.884	0.868
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	SFT + DiaTool-DPO	0.743	0.871	0.833	0.826	0.765	0.818
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	SFT + DiaTool-DPO	0.857	0.929	0.917	0.913	0.905	0.904

Source: FunctionChat-Bench evaluation (arXiv:2504.02882). Bold = best per model.

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO



Slot Accuracy

0.639 → 0.917

+43.5% relative gain

Relevance

0.826 → 0.913

+10.5% relative gain

Approaches near GPT-4o performance: 94.8% information gathering, 91% tool call rejection.

Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

1

MDP formulation enables systematic preference pairing

Defining 5 internal states and 3 query types allows automatic generation of chosen/rejected trajectory pairs without human annotation.

2

Automatic dataset construction eliminates labeling cost

Each pair teaches a specific dialogue-control lesson—redundant slot-filling, slot hallucination, or tool rejection.

3

DPO complements SFT; it does not replace it

DiaTool-DPO-only performs poorly across all models, confirming that preference optimization builds on—not supersedes—supervised training.

4

Slot-filling and tool rejection see the largest gains

LLaMA-3-8B improves Slot accuracy by +43.5% and Relevance by +10.5%, reaching near-GPT-4o conversational performance.

5

Language-agnostic and production-ready

Primary experiments in Korean generalize to English, opening broad applicability for multilingual production TA-LLMs.